

Computer Vision Project — Final Report

Helmet, Bike & Passenger Detection Using YOLOv8n

Internship: AIRI Team PITB — AI Internship Task 1

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Platform: Google Colab

Framework: Ultralytics YOLOv8

Important project-status note: The supplied notebook shows that Google Drive mounting failed and the dataset-transfer step copied zero files. The notebook then generated a small synthetic fallback dataset. Therefore, this report describes the actual notebook execution and does not claim that the full Roboflow traffic dataset was successfully trained.

1. Project Overview

This project implements an object-detection pipeline using the YOLOv8n model. The intended application is traffic-safety analysis through the detection of three classes: **Bike**, **Helmet**, and **Passenger**. The workflow includes environment setup, dataset preparation, dataset inspection, model training, test-set evaluation, inference, and an attempt at error analysis.

2. Objectives

- Prepare a YOLO-format dataset for object detection.
- Train a YOLOv8n model using pretrained weights.
- Evaluate the model on a test split using precision, recall, mAP@0.5, and mAP@0.5:0.95.
- Run inference on unseen test images.
- Identify limitations and possible improvements.

3. Tools and Technologies

Python, Google Colab, Google Drive, Ultralytics YOLOv8, PyTorch, OpenCV, Matplotlib, Seaborn, Pandas, Pillow, PyYAML, and Jupyter Notebook.

4. Dataset and Preparation

The notebook was configured to read data from `/content/drive/MyDrive/archive`. The expected source folders were `train`, `valid`, and `test`, each containing image and label folders. However, the notebook output recorded **zero files copied from Google Drive**. As a fallback, the notebook generated a synthetic dataset.

Split	Images	Label files	Source
Train	20	20	Synthetic fallback
Validation	10	10	Synthetic fallback
Test	15	15	Synthetic fallback
Total	45	45	Synthetic fallback

Classes configured in data.yaml: Bike (class 0), Helmet (class 1), Passenger (class 2).

Dataset limitation: The synthetic fallback data is not a valid substitute for a real, manually reviewed traffic dataset. The full dataset requirements of the internship task were therefore not met in this execution.

5. Model Architecture

The project uses **YOLOv8n**, the nano version of the Ultralytics YOLOv8 object-detection family. The model was initialized with pretrained weights from `yolo8n.pt` and adapted to three target classes. The model summary printed approximately **3,006,233 parameters** and **8.1 GFLOPs**.

6. Training Configuration

Parameter	Actual notebook value
Base model	yolov8n.pt
Epochs	30
Image size	640 × 640
Batch size	16
Optimizer	Auto-selected by Ultralytics
Device	Tesla T4 GPU
Framework version	Ultralytics 8.4.154
Training output folder	/content/cv_project/outputs/training_results

7. Model Evaluation

The notebook successfully evaluated the model on the test split containing 15 images and 31 object instances. The recorded results were:

Metric	Test result
Precision	0.0038
Recall	0.5347
mAP@0.5	0.0248
mAP@0.5:0.95	0.0061

Per-class test results:

Class	Images	Instances	Precision	Recall	mAP@0.5	mAP@0.5:0.95
All	15	31	0.00376	0.535	0.0248	0.00614
Bike	7	9	0.00443	0.667	0.0274	0.00643
Helmet	9	16	0.00518	0.438	0.0400	0.00979
Passenger	6	6	0.00167	0.500	0.00716	0.00221

These results are very weak overall, especially the precision and mAP values. They should be interpreted in the context of the synthetic fallback dataset and the absence of the intended real-world training data.

8. Inference Results

The notebook attempted inference on the first 15 test images with a confidence threshold of 0.25. The execution log reported **no detections for all 15 images**. Prediction outputs were configured to be saved under /content/cv_project/outputs/predictions.

9. Error Analysis

The notebook attempted to collect up to 10 low-confidence predictions using a confidence threshold of 0.10 and then filtering boxes below 0.40. The output reported **0 potential low-confidence samples saved**. Consequently, a verified table of 10 incorrect predictions was not produced.

10. Limitations

- Google Drive mounting failed during the notebook run.
- No real dataset files were copied from the configured Drive location.
- A synthetic fallback dataset was used instead of the intended traffic dataset.
- The executed dataset contained only 45 images: 20 train, 10 validation, and 15 test.

- The model produced no detections during the recorded 15-image inference run.
- The error-analysis step produced zero saved weak-prediction samples.
- The results are not suitable for claiming production-ready helmet detection performance.

11. Recommended Improvements

- Reconnect Google Drive successfully and load the original Roboflow dataset.
- Verify that every image has a matching YOLO label file.
- Use a properly reviewed and diverse dataset with the required train/validation/test split.
- Check class balance and inspect bounding boxes before training.
- Train YOLOv8n on the real dataset for at least 30 epochs.
- Save and review confusion matrices, loss curves, and per-class metrics.
- Run inference on at least 15 genuine unseen images and manually document at least 10 errors.
- Consider YOLOv8s or additional data augmentation after establishing a reliable baseline.

12. Conclusion

The notebook demonstrates the basic structure of an end-to-end YOLOv8 object-detection workflow, including project-folder creation, dependency installation, dataset configuration, training, evaluation, inference, and documentation generation. However, because the intended dataset was not loaded and a synthetic fallback dataset was used, the current execution should be treated as a pipeline test rather than a completed real-world helmet-detection system. The next priority is to successfully load and validate the actual annotated traffic dataset and repeat the full experiment.

13. Final Submission Status

Requirement	Status based on supplied notebook
YOLOv8n model	Completed
30 training epochs configured	Completed
Google Colab/Tesla T4 execution	Recorded
Real dataset loaded	Not completed
Minimum real dataset size	Not met in this run
Test inference on 15 images	Attempted; no detections recorded
Evaluation metrics	Available and reported above
10-case error analysis	Not completed
Final report	Prepared

Prepared from the supplied *HELMET-DETECTION.ipynb* notebook and the uploaded report. Date: September 17, 2026.